

Design Smartly With ALTium 2017

Altium is one of the most popular design software worldwide. With its large component libraries and smart features, it outshines all its rival software. Year after year, the software keeps updating. Here we focus on the latest version, Altium 2017.

What is new

Altium is an all-in-one comprehensive software with a seamless exchange of design data. Be it schematic capture to PCB layout, or viewing MCAD constraints to providing a natural electronics workflow, Altium does it all. It adopts a much more efficient approach with user-defined enhancement.

Object-specific keep-outs

Designers face a lot of problems in routing and spacing between components. For instance, if high-voltage paths are not taken care of during design, these can cause damage to the circuit. Designers need to ensure that these have enough creepage and clearance distance from other tracks and components.

To explain, let's consider the case of an LED driver manufacturing firm, where a designer designed a PCB without keeping in mind voltage differences across each point. During high-voltage test, the circuit failed with a

high noise and copper peeling off.

Altium 2017 helps to overcome this issue by pointing out high-voltage areas or prohibited zones known as keep-out areas during designing. Besides, it has upgraded PCB routing.

Active routing

Whether the layout is single-layered, double-layered or multilayered, routing is now easy with Altium 2017's active routing feature. From a single track to multiple tracks, all can be managed with a single click. Just select the nets and map-out the intended routing area. The tool not only saves time but also increases

the efficiency greatly. Designers can produce multilayer boards with high-class routing.

Track glossing

The glossy feature lets designers clean up the routing segments and areas, giving the best finish to the end-product. It acts like an optimisation tool for quality and length of every net on the PCB, allowing automated alignment of routing paths.

Dynamic components/net selection

Designers can move, cut, paste or select any object in the layout easily.

Earlier, during these functions, all the components used to get selected, whether required or not.

Dynamic copper

This feature helps designers to maintain the copper volume around components, holes and vias as per the standard value of manufacturing. It easily combines copper shapes as per the layout requirement. Thus, judicious use of copper helps designers and manufacturers to save costs on copper usage.

Back drilling

Signal attenuation, crosstalks and reflections are the major drawbacks of any design

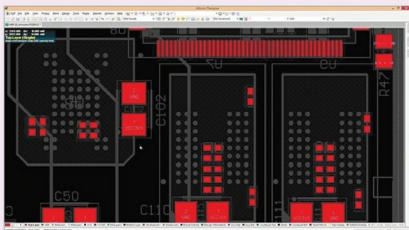


Fig. 1: Object-specific keep-outs (Image courtesy: <https://i.ytimg.com>)

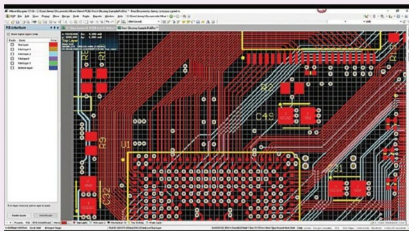


Fig. 2: Active routing (Image courtesy: <https://i.ytimg.com>)